Magnetostatics

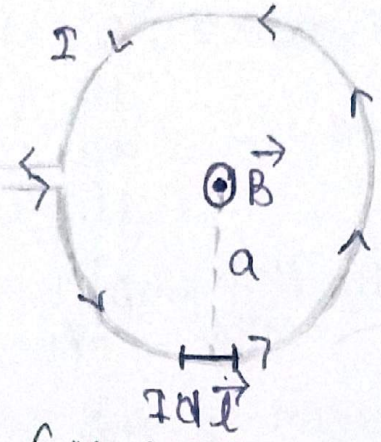
(01) Bio-savart law

$$d\vec{B} = \left(\frac{\mu_0}{4\pi}\right)\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

$$\vec{B} = \int d\vec{B} \Rightarrow \vec{B} = \frac{\mu_0}{4\pi}\left(\frac{I\,d\vec{l}\times\hat{r}}{r^2}\right)$$

01 Magnetic field intensity due to a current carrying loop

Derivation 01



$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

since $d\vec{l}$ is $\perp$ to $\vec{r}$ ($\hat{r}$)

$$|d\vec{l}\times\hat{r}| = |d\vec{l}|\,|\hat{r}|\sin\theta$$
$$= dl \cdot 1 \cdot 1$$
$$= dl$$

$$dB = \frac{\mu_0}{4\pi}\left(\frac{I\,dl}{r^2}\right)$$

$$B = \oint dB = \oint \frac{\mu_0}{4\pi}\left(\frac{I\,dl}{r^2}\right)$$

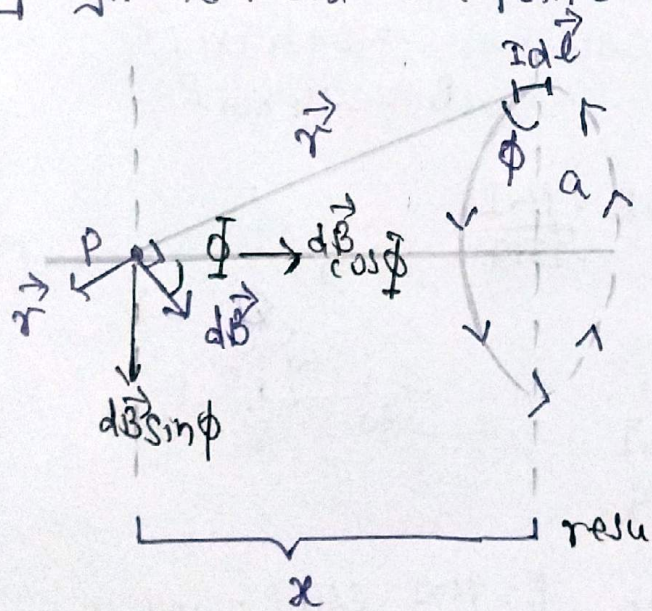
$$= \frac{\mu_0}{4\pi}\frac{I}{r^2}(2\pi r) \Rightarrow B = \frac{\mu_0 I}{2r}$$

Here $r = a$

$$B = \frac{\mu_0 I}{2a}$$

02 Magnetic field at a point P on an axis of a current loop

Derivation 02



$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

since $d\vec{l} \perp \vec{r}$

$$|d\vec{l}\times\hat{r}| = |d\vec{l}|\,|\hat{r}|\sin\theta$$
$$= dl \cdot 1 \cdot 1$$

$$B = \oint dB\cos\phi \qquad \left(\cos\phi = \frac{a}{r}\right)$$

$$= \oint \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}\cos\phi$$

resultant forces from $dB\sin\phi$ cancel out

$$B = \frac{\mu_0}{4\pi}\frac{I}{r^2}\left(\frac{a}{r}\right)2\pi a$$

$$B = \frac{\mu_0}{4\pi}\frac{I}{r^3}2\pi a^2 = \frac{\mu_0 I a^2}{2r^3}$$

$$B = \frac{\mu_0 I a^2}{2(a^2+x^2)^{3/2}}$$

- $x \gg a \Rightarrow B = \frac{\mu_0 I a^2}{2x^3}$
- $x = 0 \Rightarrow B = \frac{\mu_0 I}{2a}$ (At the center)

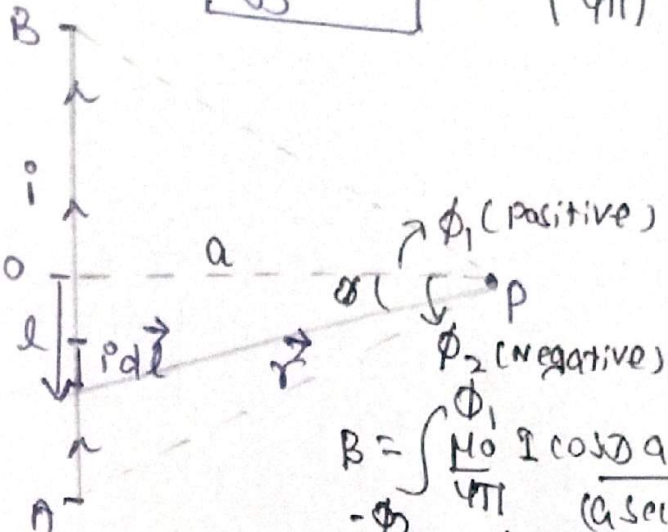
03 Magnetic field due to a finite straight current carrying conductor

Derivation 03

$$d\vec{B} = \left(\frac{\mu_0}{4\pi}\right)\frac{I\,d\vec{l}\times\hat{r}}{r^2} = \left(\frac{\mu_0}{4\pi}\right)\frac{I\,|d\vec{l}|\,|\hat{r}|\sin\theta}{r^2}$$

$$= \frac{\mu_0}{4\pi}\frac{I\cos\theta\, dl}{r^2} \qquad (\cos\theta = \frac{a}{r})$$

$$B = \int_A^B \frac{\mu_0}{4\pi}\frac{I\cos\theta\, dl}{r^2}$$



$r = a\sec\theta$, $l = a\tan\theta$

$\frac{dl}{d\theta} = a\sec^2\theta$

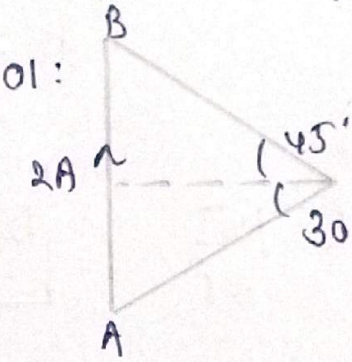
$dl = a\sec^2\theta\, d\theta$

$$B = \int_{-\phi_2}^{\phi_1} \frac{\mu_0}{4\pi}\frac{I\cos\theta\, a\sec^2\theta\, d\theta}{(a\sec\theta)^2}$$

$$B = \frac{\mu_0}{4\pi}\int_{-\phi_2}^{\phi_1}\frac{\cos\theta}{a}\,d\theta \Rightarrow B = \frac{\mu_0 I}{4\pi a}\sin\theta\Big|_{-\phi_2}^{\phi_1}$$

$$\boxed{B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)}$$

Eg01:



$$B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$$

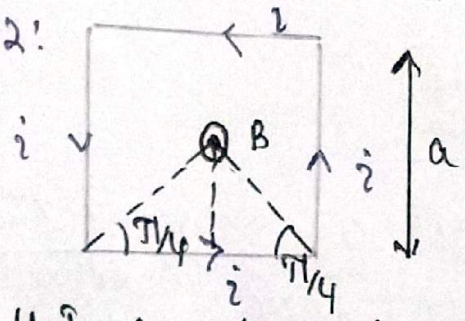
$$= \frac{\mu_0 I}{4\pi a}\left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right)$$

$$= \frac{10^{-7}}{10}(2+\sqrt{2})$$

$$= (2+\sqrt{2})\times 10^{-8}$$

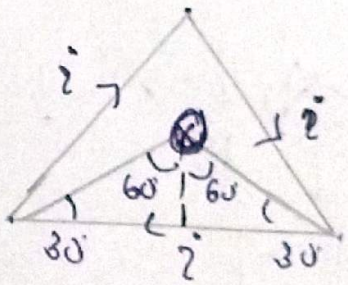
$$B \simeq 3.14\times 10^{-8}$$

Eg02:



$$B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$$

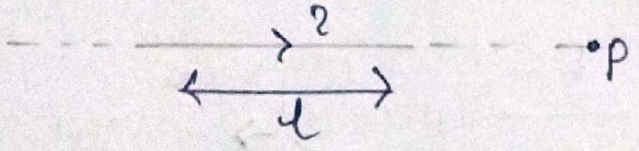
$$= \frac{\mu_0 I}{4\pi(a/2)}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = \frac{\mu_0 I}{2\pi a}\cdot\frac{2}{\sqrt{2}} \Rightarrow B = \frac{\mu_0 I}{\sqrt{2}\pi a}$$



Eg03: $B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$

$$= \frac{\mu_0 I}{4\pi a}\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4}\frac{\mu_0 I}{\pi a}$$

Eg04:



$$B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$$

$\phi_1 = \phi_2 = 0$

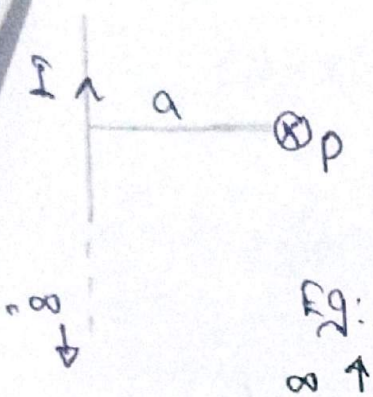
$B = \frac{\mu_0 I}{4\pi a}(0+0) = 0 \Rightarrow B = 0$

Magnetic field intensity due to a infinitely long conductor

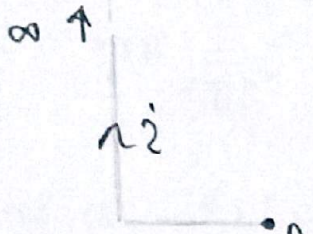
$$B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$$

$$\phi_1 = \pi/2 = \phi_2 \Rightarrow B = \frac{\mu_0 I}{4\pi a}(\sin\pi/2 + \sin\pi/2)$$

$$B = 2\frac{\mu_0 I}{4\pi a} \Rightarrow \boxed{B = \frac{\mu_0 I}{2\pi a}}$$



Eg: Semi-infinite wire



$$B = \frac{\mu_0 I}{4\pi a}(\sin\phi_1 + \sin\phi_2)$$

$$= \frac{\mu_0 I}{4\pi a}(1\sin\pi/2 + 0) = \frac{\mu_0 I}{4\pi a}$$

$$B = \frac{\mu_0 I}{4\pi a}$$

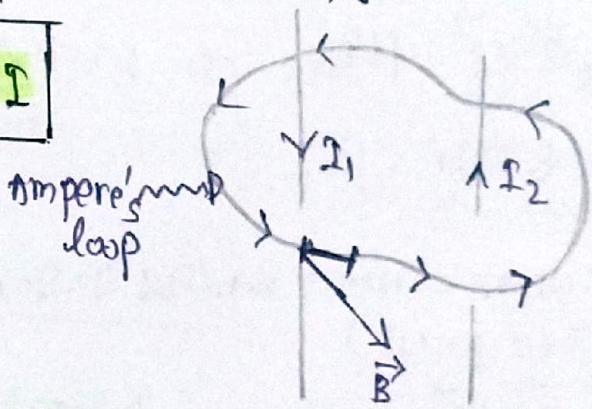
(02) Ampere's law in magnetic field

- The line integral of $\vec{B}.d\vec{l}$ around any closed path equals to $\mu_0 I$ where I is the total steady current passing through the closed path

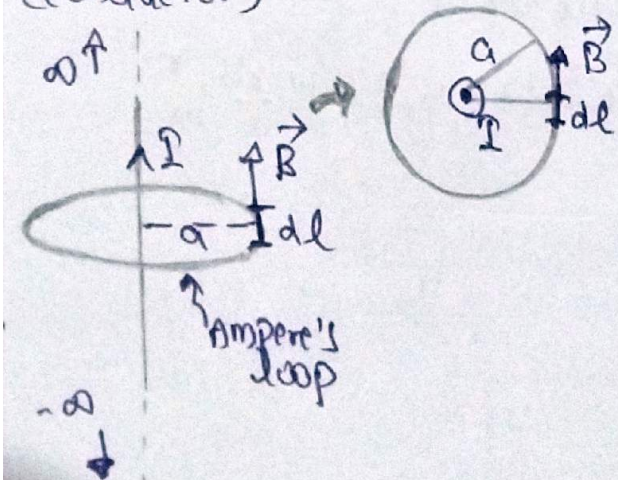
$$\boxed{\oint_l \vec{B}.d\vec{l} = \mu_0 I}$$

- useful for calculating mag fields due to symmetrical current flow

$$\left(\oint \vec{E}.d\vec{A} = \frac{\sum q}{\epsilon_0}\right) \leftarrow \text{similar}$$



[01] Magnetic field due to infinitely long current carrying wire (conductor)



$$\oint_l \vec{B}.d\vec{l} = \oint_l B.dl\cos 0 = \mu_0 I$$

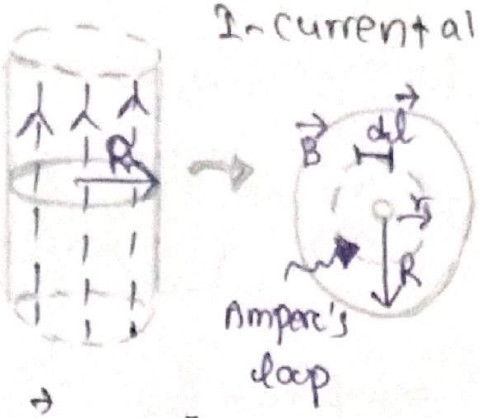
$$B\oint_l dl = \mu_0 I$$

$$B(2\pi a) = \mu_0 I$$

$$\boxed{B = \frac{\mu_0 I}{2\pi a}}$$

02 Magnetic field due to infinitely long thick current conductor

R - radius of the conductor
I - current along the conductor



(a) Inside point ($r < R$)

current density

$$J = \frac{I}{A} = \frac{I}{\pi R^2}$$

$$i = J\pi r^2$$

$$i = \frac{I}{\pi R^2}\pi r^2 = \frac{I r^2}{R^2}$$

$$\oint_l \vec{B}\cdot d\vec{l} = \mu_0 i$$

$$\oint_l B\, dl\cos\theta = \mu_0 \frac{I r^2}{R^2} \quad \rightarrow \quad B(2\pi r) = \mu_0 \frac{I r^2}{R^2}$$

$$B\oint_l dl = \mu_0 \frac{I r^2}{R^2}$$

$$B = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$$

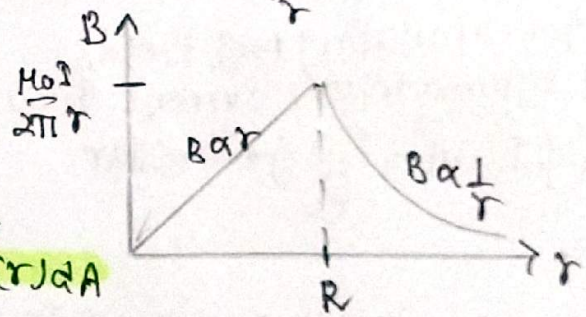
So in an inside point $B \propto r$

(b) outside point ($r > R$)

$$\oint_l \vec{B}\cdot d\vec{l} = \mu I_{enc} \quad \rightarrow \quad B = \left(\frac{\mu_0 I}{2\pi}\right)\frac{1}{r}$$

$$B\,2\pi r = \mu_0 I$$

So in an outside point $B \propto \frac{1}{r}$



Eg: when current density $J = J_0 r$ (Inside point)

$$\oint_l \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0\left(2\pi J_0 \frac{r^3}{3}\right)$$

$$B = \frac{\mu_0 J_0 r^2}{3}$$

$$I_{enc} = \int_0^r J(r)\,dA$$

$$= \int_0^r (J_0 r')(2\pi r' dr')$$

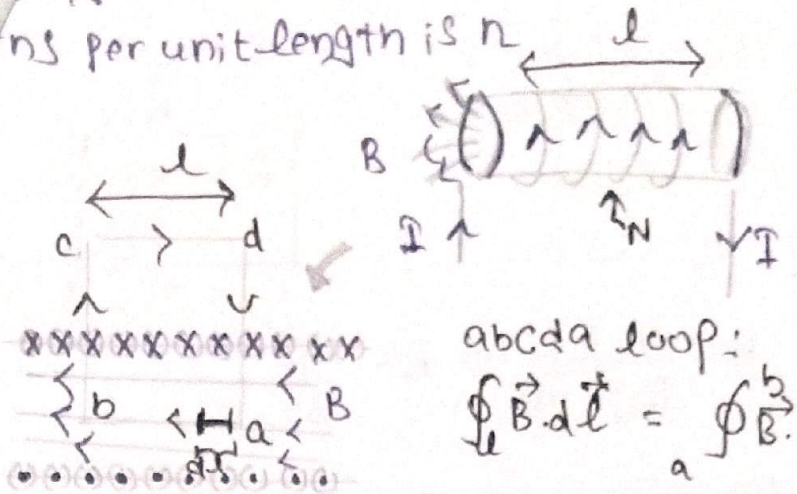
$$= 2\pi J_0 \int_0^r r'^2 dr' = 2\pi J_0 \frac{r'^3}{3}\Big|_0^r$$

$$= \frac{2\pi J_0 r^3}{3}$$

(outside point)

$$I_{total} = \frac{2\pi J_0 R^3}{3} \quad \rightarrow \quad B = \frac{\mu_0 J_0 R^3}{3r}$$

Magnetic field of a solenoid carrying a current I. Number of turns per unit length is n



abcda loop:

$$\oint_l \vec{B}\cdot d\vec{l} = \int_a^b \vec{B}\cdot d\vec{l} + \int_b^c \vec{B}\cdot d\vec{l} + \int_c^d \vec{B}\cdot d\vec{l} + \int_d^a \vec{B}\cdot d\vec{l}$$

$$= \int_a^b B\,dl\cos 0 + \int_b^c B\cdot dl\cos 90^\circ + \int_c^d 0\cdot dl\cos 0 + \int_d^a B\,dl\cos 90^\circ$$

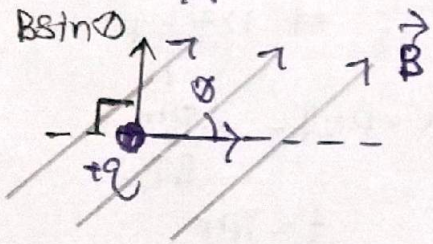
So:

$$\oint_l \vec{B}\cdot d\vec{l} = B\int_a^b dl = \mu_0 I_{enc} \qquad I_{net} = NI = nlI$$

$$\mu_0 nIl = Bl$$

$$\boxed{B = \mu_0 nI}$$

(03) Force on charge in a magnetic field



$$\boxed{F_B = BqV\sin\theta}$$

Direction: Flemings Left hand rule.

$$\boxed{\vec{F}_B = q(\vec{V}\times\vec{B})}$$

In a electromagnetic field force exerted on a moving charge:

$$\boxed{\vec{F} = q\vec{E} + q(\vec{V}\times\vec{B})}$$ "Lorentz force"

$\vec{F} \perp \vec{V}$ & $\vec{F} \perp \vec{B}$

work done by magnetic force:

$$W = \vec{F}\cdot d\vec{s}$$

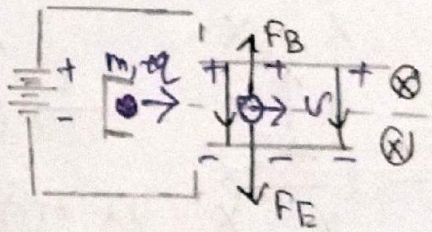
$$W = |\vec{F}||d\vec{s}|\cos 90^\circ = 0$$

$\downarrow \vec{F}_E = qE \uparrow F_B = BqV$

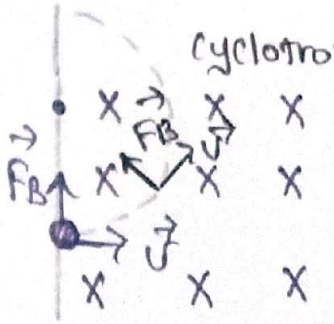
$qE = BqV$

$E = BV$ ⇒ $\boxed{V = \frac{E}{B}}$

velocity selector



01 Equations of a charge moving in a mag. field



cyclotron

$F_B = \frac{mv^2}{R}$ $\quad \vec{F_B} = q(\vec{v} \times \vec{B})$ $\quad qvB = \frac{mv^2}{R}$

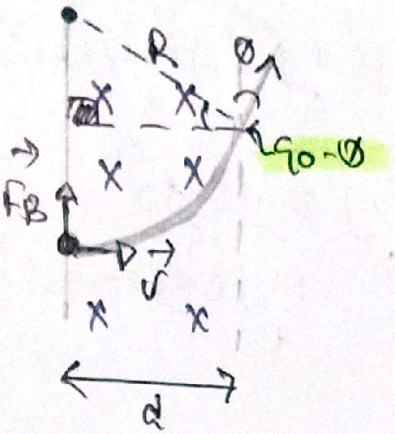
$F_B = qvB\sin 90$

$= qvB$

$\boxed{R = \frac{mv}{Bq}}$

$E = \frac{1}{2}mv^2$

$v = \sqrt{\frac{2E}{m}}$ $\quad R = \frac{m}{Bq}\sqrt{\frac{2E}{m}} \Rightarrow R = \frac{\sqrt{2Em}}{Bq}$



$d/R = \cos(90-\theta)$

$R\sin\theta = d$ $\quad \Rightarrow \frac{mv}{Bq}\sin\theta = d$

$\boxed{\sin\theta = \frac{Bqd}{mv}}$

$T = \frac{2\pi R}{v}$

$\boxed{T = \frac{2\pi m}{Bq}} \Rightarrow \boxed{f = \frac{Bq}{2\pi m}}$ ← cyclotron frequency

Eg: when (a) $d = \frac{mv}{2Bq}$ $\quad R = \frac{mv}{Bq} \Rightarrow d = \frac{R}{2}$ $\quad R\sin\theta = d$ $\quad \sin\theta = 1/2$

$R\sin\theta = R/2$ $\quad \theta = 30°$

(b) $d = \frac{mv}{Bq}$ $\quad \sin\theta = 1$

$t = \left(\frac{\theta}{2\pi}\right)T = \left(\frac{\pi/6}{2\pi}\right)T = \frac{T}{12} \Rightarrow 12t = T$

$d = R$ $\quad \theta = \pi/2$

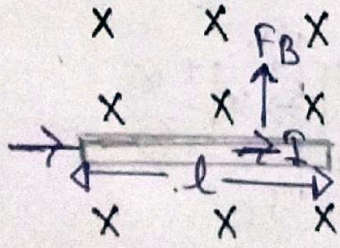
$T = \frac{2\pi}{v}\left(\frac{mv}{Bq}\right) = \frac{2\pi m}{12q} \Rightarrow 12t = \frac{2\pi m}{Bq}$

$t = \frac{\pi m}{2Bq}$

$t = \frac{\pi m}{6Bq}$

(c) $d = \frac{2mv}{Bq}$ $\quad d = 2R$ $\quad t = T/2$ $\quad t = \frac{\pi m}{Bq}$

04 Force acting on a current conductor placed in a magnetic field



$F_B = q(\vec{v} \times \vec{B})$

force on one electron:

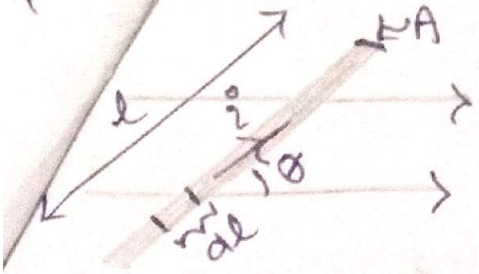
$F = e(\vec{v_d} \times \vec{B})$

v_d - drift velocity

$j = nev_d$

$\frac{I}{A} = nev_d \Rightarrow I = v_d A n e$

$|\vec{F}| = evB\sin\theta$



$F = \int dF = \int (dl \times A \times n)\, e v_d \sin\theta\, B$

$= \int_0^l (nAe v_d B \sin\theta)\, dl$

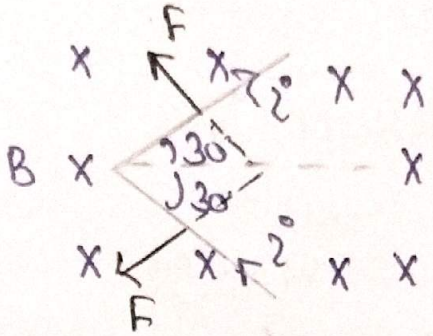
$= nAe v_d B \sin\theta \int_0^l dl$

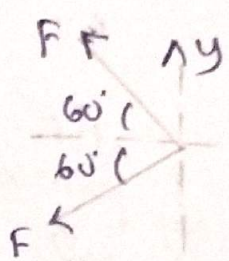
$F = nAe v_d B \sin\theta\, l \Rightarrow \boxed{F = IBl\sin\theta}$

$\boxed{F = I(\vec{l} \times \vec{B})}$

$F_{max} = BIl \quad (\theta = \pi/2)$

Eg:

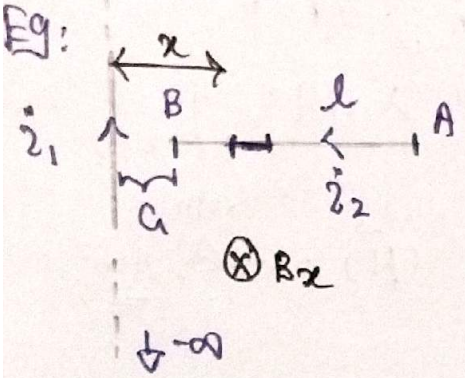




$\uparrow F_y = F\sin 60 + (-F\sin 60) = 0$

$\leftrightarrow F_x = 2F\cos 60 = F$

$F_{net} = IBl$



$dF = B_x i_2 dx$

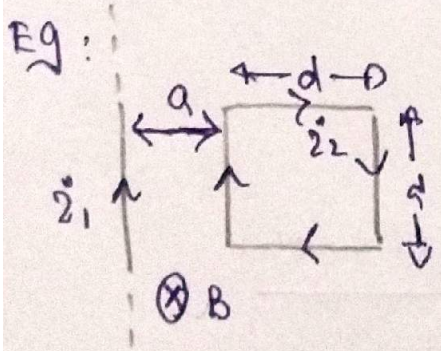
where $B_x = \frac{\mu_0 i_1}{2\pi x}$

$dF = \frac{\mu_0 i_1}{2\pi x} \cdot i_2 dx \Rightarrow F = \int_a^{a+l} \frac{\mu_0 i_1 i_2}{2\pi} \frac{dx}{x}$

$F = \frac{\mu_0 i_1 i_2}{2\pi} \ln\left(\frac{a+l}{a}\right)$

$F = \frac{\mu_0 i_1 i_2}{2\pi} \ln|x| \Big|_a^{a+l}$

Eg:



current i_2 flows in opposite directions in top and bottom so:

$F_{top} + F_{bottom} = 0$

$F_{left} = i_2 dB(a) = i_2 d\left(\frac{\mu_0 i_1}{2\pi a}\right) = \frac{\mu_0 i_1 i_2 d}{2\pi a}$

$F_{right} = i_2 dB(a+d) = i_2 d\left(\frac{\mu_0 i_1}{2\pi(a+d)}\right) = \frac{\mu_0 i_1 i_2 d}{2\pi(a+d)}$

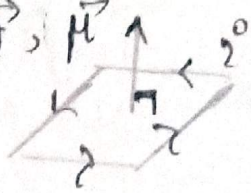
$F_{net} = F_{left} - F_{right}$

$= \frac{\mu_0 i_1 i_2 d}{2\pi a} - \frac{\mu_0 i_1 i_2 d}{2\pi(a+d)}$

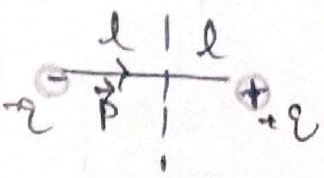
$= \frac{\mu_0 i_1 i_2}{2\pi}\left(\frac{a+d-a}{a(a+d)}\right) \Rightarrow F_{net} = \frac{\mu_0 i_1 i_2 d^2}{2\pi a(a+d)}$

(05) Magnetic moment ($\vec{\mu}$)

$\vec{A}$, $\vec{\mu}$

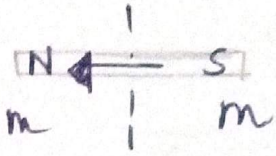


$\vec{\mu} = i\vec{A}$ (unit: Am^2) A - Area vector



$\vec{p} = q \times 2\vec{l}$

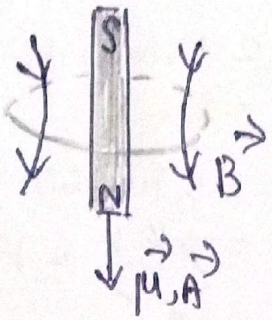
$\mu = m \times 2l$

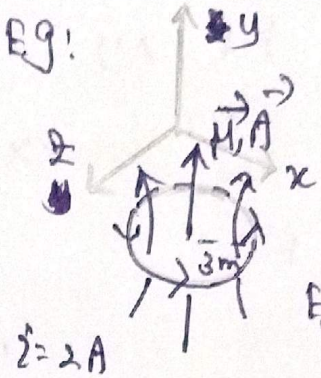


Direction: (S) to (N)

m - pole strength (magnetic strength)

Current carrying loops behave like magnets





Eg:

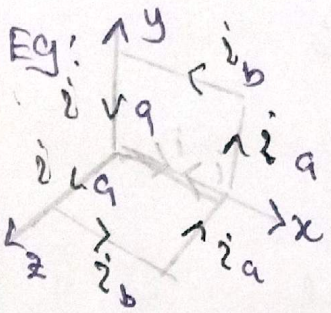
$|\vec{M}| = M = iA$

$= 2(\pi 3^2)$

$= 57.52\ Am^2$

$i = 2A$

Eg:

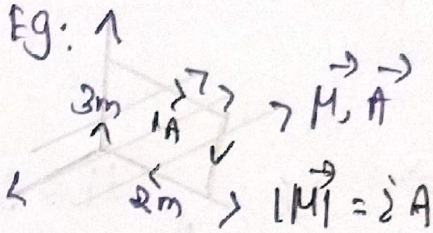


$|M_1| = i(ab)$

$|M_2| = i(ab)$

$|M_{net}| = \sqrt{2}(i ab)$

Eg:



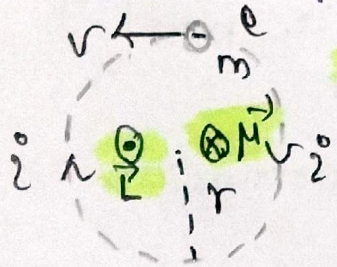
$|\vec{M}| = iA$

$= 1(2 \times 3)$

$= 6\ Am^2$

$(\vec{M}) = (6\hat{k})\ Am^2$

01 Magnetic dipole moment (μ) of a revolving electron. (Bohr Model)



$i = \frac{e}{T}$ $T = \frac{2\pi r}{v}$ $\Rightarrow i = \frac{ev}{2\pi r}$

Now $|(\vec{\mu})| = i \cdot A$

$= \frac{ev}{2\pi r} \times \pi r^2$

$\mu = \frac{evr}{2}$ (direction (S) to (N))

Since $\vec{L} = \vec{r} \times \vec{p}$

$|\vec{L}| = r \times mv \sin 90$ $\Rightarrow \frac{\mu}{L} = \frac{evr/2}{mvr} = \frac{e}{2m}$ $\Rightarrow \boxed{\mu = \frac{e}{2m} \cdot L}$

$L = mvr$

$\boxed{\vec{\mu} = -\frac{e}{2m} \cdot \vec{L}}$

from the Bohr Model $L = \frac{nh}{2\pi}$ $n = 1, 2, \ldots$

$\boxed{\mu = \frac{e}{2m} \cdot \frac{nh}{2\pi}}$ $\rightsquigarrow$ Bohr magneton (n=1) $= \boxed{\frac{eh}{4\pi m} = \mu_B}$